

Nevra: A Consumer Credit Layer for the On-Chain Economy

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Abstract

Nevra is an undercollateralized consumer credit protocol issuing real-time stablecoins credit lines underwritten against verifiable on-chain and offchain creditworthiness rather than posted collateral. Borrowers connect a wallet and a bank account to receive a credit line priced on expected loss, while suppliers deposit stablecoins into a two-tranche pool that earns a risk-adjusted yield. Defaults are absorbed by a junior tranche and recovered through licensed U.S. collections agencies bidding on-chain on non-performing debt. By pricing credit to individuals the way existing money markets price collateral, Nevra extends the missing primitive of on-chain finance: unsecured consumer credit at scale.

Beyond protocol mechanics, Nevra targets a deeper mission: to reshape the foundations of TradFi consumer credit. The current system relies on opaque scoring, slow onboarding, and high APRs that disproportionately burden households. Nevra replaces this with transparent, on-chain underwriting, instant disbursement, and risk-aligned pricing. Over time, Nevra aims to become the core credit layer connecting individuals, businesses, and autonomous agents to the on-chain economy, driving long-lasting change in how credit is rated, distributed, and recovered across both crypto and traditional finance.

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1 Why Credit Remains Unsolved On-Chain

On-chain finance has solved the medium of exchange and never solved credit. The entire lending stack is overcollateralized: a user must already hold the asset they want to borrow against. That excludes the majority of economic actors, for whom credit is not a function of assets held but of income, history, and willingness to repay.

The market this leaves untouched is enormous. U.S. household debt stands at roughly \$18.8 trillion, carried at an average APR near 21%.¹ Traditional issuers underwrite this market profitably but slowly, behind opaque models and weeks of onboarding. On-chain rails can underwrite and disburse the same credit in minutes, and price it transparently, yet no protocol today does so for individuals.

Nevra closes that gap. It treats a borrower’s full financial picture, on-chain and off, as the collateral, and prices a credit line against the probability that the borrower repays.

More broadly, Nevra seeks to challenge the TradFi credit paradigm itself. The current system concentrates power in a few scoring agencies and lenders, uses black-box models, and often prices risk in ways that hurt consumers rather than reflect true expected loss. Nevra proposes a new foundation: credit identity bound to the individual, underwriting driven by verifiable data, and recovery mechanisms that are transparent and auditable. This is not just a protocol upgrade but a structural shift toward a more open, fair, and efficient credit system.

2 System Overview

Nevra is a peer-to-pool credit market with three primary functions:

1. **Supply pool:** suppliers deposit USDC into a two-tranche pool. The senior tranche earns a stable yield with first claim on repayments; the junior tranche earns a levered yield and absorbs first losses on default.
2. **Underwriting engine:** an offchain scoring engine ingests a borrower’s on-chain history and offchain credit data, returns a credit line size, a default risk premium, and a repayment schedule, and posts the terms on-chain with a zero-knowledge proof of the underlying data.
3. **Recovery module:** on default, the protocol slashes the borrower’s Nevra credit score and opens an on-chain auction where licensed U.S. collections agencies bid to recover the debt on contingency.

Idle pool liquidity is routed into an established on-chain money market to earn a base yield until drawn down by borrowers. The protocol is built to extend across chains as liquidity and demand justify.

The long-term vision extends far beyond consumer credit lines. Nevra aims to become the universal credit layer for the on-chain economy, supporting:

¹Federal Reserve Bank of New York, Household Debt and Credit Report. Average APR per the Federal Reserve G.19 consumer credit release.

1. Pure cash-flow underwriting for individuals, where income and revenue rather than holdings determine creditworthiness.
2. Business credit for startups, DAOs, and small enterprises that can prove programmatic cash flows.
3. Credit for autonomous agents and AI systems that can commit to obligation contracts and repay via on-chain revenue streams.

In this future, Nevra does not just add a new lending product. It redefines how credit is created, priced, and enforced across both crypto and TradFi, moving the entire system toward transparency, fairness, and efficiency.

3 Credit Underwriting

Each borrower is scored against a Nevra credit identity: a Soulbound token carrying a zero-knowledge proof of the borrower’s credit data. The engine prices every credit line on its expected loss,

$$EL = PD \times LGD \times EAD, \tag{1}$$

where PD is the probability of default over the term, LGD is the loss given default net of expected recovery, and EAD is the exposure at default. The default risk premium charged to the borrower is derived directly from EL, so that the pool is compensated for the risk it underwrites:

$$APR_{\text{borrower}} = APR_{\text{base}} + \text{DRP}(EL). \tag{2}$$

Credit lines are bounded below by a minimum borrow floor. Below this floor a defaulted balance is uneconomic to pursue, since collections agencies carry a fixed cost per dossier and decline files beneath it. The floor keeps every issued line inside the recoverable range.

This underwriting model is designed to be portable across systems. As Nevra integrates with more off-chain data sources and TradFi partnerships, the same expected-loss framework can be used to rate consumer loans, small business loans, and even structured credit products. The goal is to replace opaque scoring with a single, transparent, and auditable credit model that both on-chain and off-chain lenders can rely on.

4 Loss Waterfall and Recovery

Losses flow through the tranches in strict order. For a realized loss ℓ on the pool, the junior tranche absorbs up to its full balance J before any loss reaches the senior tranche:

$$\text{Loss}_{\text{junior}} = \min(\ell, J), \quad \text{Loss}_{\text{senior}} = \max(0, \ell - J). \tag{3}$$

A delinquent line is not written down to zero on day one. Its carrying value reflects the probability of recovery, which decays over time as collection prospects fade. On default the protocol opens a Dutch auction: the contingency share offered to collections agencies rises until one accepts, at which point the agency pursues recovery through skip tracing, credit furnishing, and legal remedy. Recovered principal flows back to the pool net of the agency's contingency share.

Because the default and its terms are committed on-chain and immutable, the cost of strategic default rises relative to an offchain creditor. This tightens spreads and reduces the credit rationing that forces overcollateralization everywhere else on-chain.

The recovery module is also a cornerstone of Nevra's broader vision. By placing debt auctions and recovery terms on-chain, Nevra creates a transparent market for non-performing debt that can be audited by anyone. This contrasts sharply with the opaque secondary markets for distressed debt in TradFi, where terms are hidden and power is concentrated. In the long run, Nevra aims to make this on-chain recovery market a standard that TradFi itself adopts, further driving systemic change.

5 Vision and Long-Term Impact

Nevra is not only a protocol for undercollateralized loans. It is an attempt to rebuild the foundations of consumer credit. The current TradFi system is built on:

1. Opaque scoring models that individuals cannot inspect or challenge.
2. Slow onboarding that takes weeks and requires extensive paperwork.
3. High APRs that often reflect operational inefficiencies and rent-seeking rather than true expected loss.
4. Recovery processes that are hidden from the public and hard to audit.

Nevra proposes a different path:

1. Transparent underwriting based on verifiable on-chain and off-chain data.
2. Instant credit disbursement once identity and data are verified.
3. Risk-aligned pricing where APR reflects expected loss and recovery prospects rather than arbitrary margins.
4. On-chain recovery markets where debt auctions and outcomes are public and auditable.

Over time, Nevra aims to shift the entire credit ecosystem toward this model. The protocol will start with consumer credit lines for individuals who can prove assets and history, then evolve toward pure cash-flow underwriting. As integrations with banks, data providers, and potentially TradFi lenders grow, Nevra can become the shared credit layer that both crypto protocols and traditional institutions use to rate and price loans.

This is the long-lasting change Nevra seeks: a credit system where power is distributed, data is verifiable, pricing is transparent, and recovery is open. If successful, Nevra does not just add a new product to the market. It changes how credit works for individuals, businesses, and autonomous agents across the global economy.

References

[0] Federal Reserve Bank of New York, *Household Debt and Credit Report*.

[0] Federal Reserve, *G.19 Consumer Credit Release*.